

One-Way Sensitivity Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Overview

This one-way sensitivity analysis varies economic parameters by the deterministic sensitivity-analysis range and summarizes the impact on net monetary benefit (NMB). The analysis uses the single joint economic model and the economic strategies returned by `get_strategies()`.

2 Parameter Ranges

Table 1: One-way sensitivity-analysis parameter ranges

	Parameter	Low	High
c_drug_nivo	c_drug_nivo	EUR 11,138	EUR 16,708
c_drug_FLOX	c_drug_FLOX	EUR 342	EUR 512
c_test_CRP	c_test_CRP	EUR 13	EUR 19
c_test_NGS	c_test_NGS	EUR 2,014	EUR 3,022
c_test_CT	c_test_CT	EUR 309	EUR 463
c_test_blood	c_test_blood	EUR 13	EUR 19
c_other_visit	c_other_visit	EUR 26	EUR 40
c_other_baseline	c_other_baseline	EUR 424	EUR 636
c_other_follow	c_other_follow	EUR 26	EUR 40
c_other_last	c_other_last	EUR 11,042	EUR 16,564
u_np	u_np	0.584	0.876
u_p	u_p	0.472	0.708
p_crp	p_crp	0.270	0.406
p_tmb_braf	p_tmb_braf	0.359	0.539

3 Base Case and Optimal Strategy

Table 2: Base case at WTP = EUR 51,000

Strategy	Cost	QALYs	NMB
Standard of Care	EUR 21,056	1.350	EUR 47,783
CRP-guided	EUR 53,791	1.389	EUR 17,026
TMB/BRAF-guided	EUR 63,235	1.351	EUR 5,684

4 Tornado Plots

Creating tornado plot for strategy: control

Creating tornado plot for strategy: crp

Creating tornado plot for strategy: tmb_braf

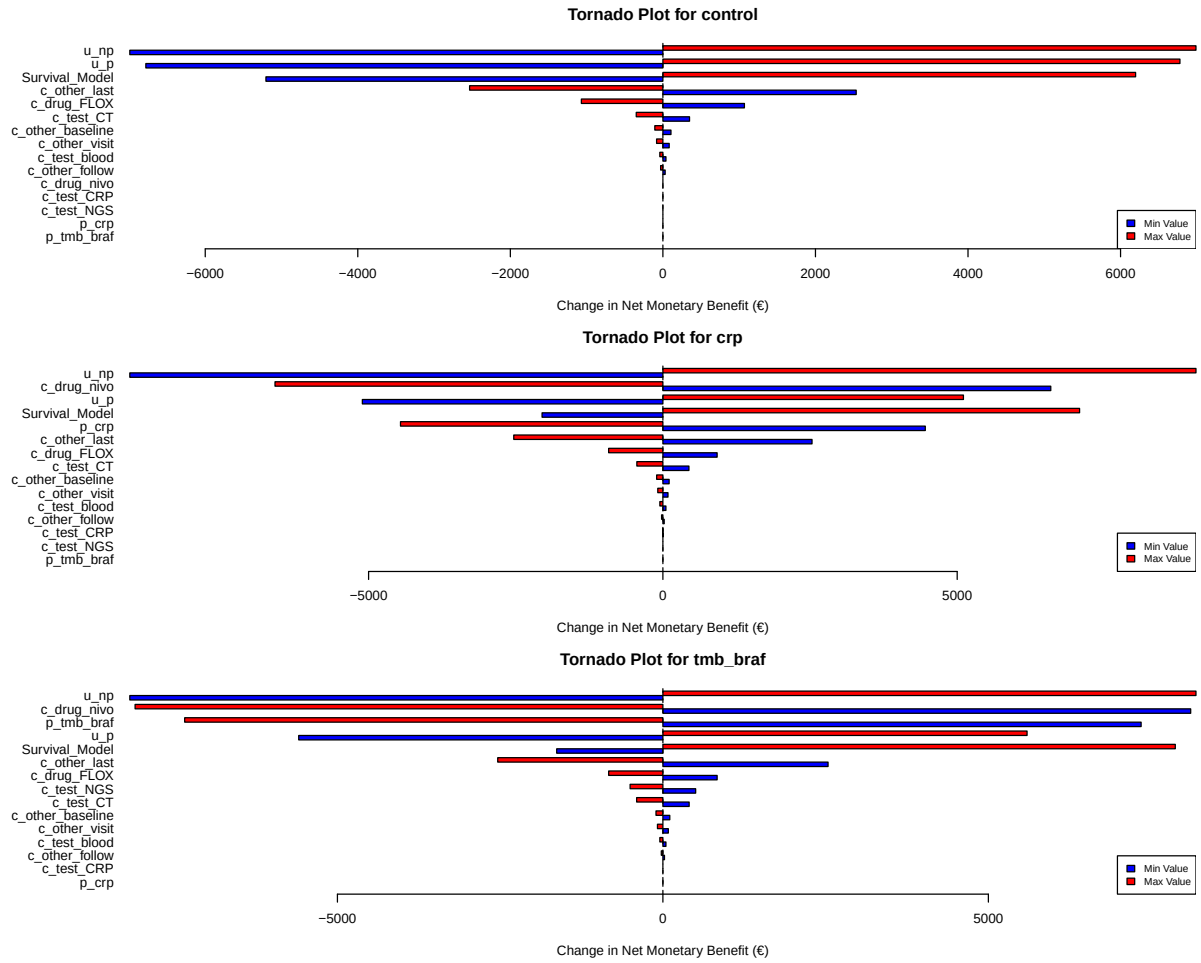


Figure 1: Tornado plots by economic strategy

5 Parameter Impact Ranking

Table 3: Top one-way sensitivity drivers by strategy

Strategy	Rank	Parameter	NMB Range
Standard of Care	1	u_np	EUR 13,978
Standard of Care	2	u_p	EUR 13,558
Standard of Care	3	Survival_Model	EUR 11,402
CRP-guided	1	u_np	EUR 18,114
CRP-guided	2	c_drug_nivo	EUR 13,180
CRP-guided	3	u_p	EUR 10,213
TMB/BRAF-guided	1	u_np	EUR 16,379
TMB/BRAF-guided	2	c_drug_nivo	EUR 16,216
TMB/BRAF-guided	3	p_tmb_braf	EUR 14,693

Table 4: Structural sensitivity to parametric survival distribution

Strategy	Base NMB	Lowest NMB	Highest NMB	Range
Standard of Care	EUR 47,783	EUR 42,578 (lognormal)	EUR 53,980 (exponential)	EUR 11,402
CRP-guided	EUR 17,026	EUR 14,973 (gompertz)	EUR 24,105 (exponential)	EUR 9,133
TMB/BRAF-guided	EUR 5,684	EUR 4,053 (gompertz)	EUR 13,556 (exponential)	EUR 9,503

6 Interpretation

The tornado plots and rankings show which parameters drive uncertainty in NMB for the single economic model. Survival distribution sensitivity is shown separately because it reflects extrapolation uncertainty rather than a scalar model input.

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